Linear Algebra: Homework 3

Yifan Chen

2026 Fall

Problem 1 [12 pts]

Label the following statements as true or false.

(a) The zero vector space has no basis.

(b) Every vector space that is generated by a finite set has a basis.

(c) Every vector space has a finite basis.

(d) A vector space cannot have more than one basis.

(e) If a vector space has a finite basis, then the number of vectors in every basis is the same.

(f) The dimension of $P_n(\mathbb{F})$ is n.

(g) The dimension of $M_{m\times n}(\mathbb{F})$ is $m+n$.

(h) Suppose that V is a finite-dimensional vector space, that S_1 is a linearly independent subset of V, and that S_2 is a subset of V that generates V. Then S_1 cannot contain more vectors than S_2.

(i) If S generates the vector space V, then every vector in V can be written as a linear combination of vectors in S in only one way.

(j) Every subspace of a finite-dimensional space is finite-dimensional.

(k) If V is a vector space having dimension n, then V has exactly one subspace with dimension 0 and exactly one subspace with dimension n.

(l) If V is a vector space having dimension n, and if S is a subset of V with n vectors, then S is linearly independent if and only if S spans V.

Answer:

The answers are, in order, (a. to d.) F, T, F, F, (e. to h.) T, F, F, T, (i. to l.) F, T, T, T.

Problem 2 [10 pts]

Let W denote the subspace of $\mathbb{R}^5$ consisting of all the vectors having coordinates that sum to zero. The vectors

$$\begin{aligned} u_1 &= (2,-3,4,-5,2), & u_2 &= (-6,9,-12,15,-6),\\ u_3 &= (3,-2,7,-9,1), & u_4 &= (2,-8,2,-2,6),\\ u_5 &= (-1,1,2,1,-3), & u_6 &= (0,-3,-18,9,12),\\ u_7 &= (1,0,-2,3,-2), & u_8 &= (2,-1,1,-9,7) \end{aligned}$$

generate W. Find a subset of the set $\{u_1, u_2, \ldots, u_8\}$ that is a basis for W.

Answer:

We first form the vectors into a 5×8 matrix

$$\begin{bmatrix} 2 & -6 & 3 & 2 & -1 & 0 & 1 & 2 \\ -3 & 9 & -2 & -8 & 1 & -3 & 0 & -1 \\ 4 & -12 & 7 & 2 & 2 & -18 & -2 & 1 \\ -5 & 15 & -9 & -2 & 1 & 9 & 3 & -9 \\ 2 & -6 & 1 & 6 & -3 & 12 & -2 & 7 \end{bmatrix},$$

and apply the row reduction:

$$\begin{bmatrix} 1 & -3 & 3/2 & 1 & -1/2 & 0 & 1/2 & 1 \\ 0 & 0 & 1 & -2 & -1/5 & -6/5 & 3/5 & 4/5 \\ 0 & 0 & 0 & 0 & 1 & -4 & -23/21 & -19/21 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Since the row reduction does not change the linear dependency of columns, the pivot columns of the original matrix are linearly independent, and the non-pivot columns can be written as linear combinations of the pivot columns. Therefore, the vectors u_1, u_3, u_5, u_7 corresponding to the pivot columns form a basis of W.

Problem 3 [8 pts]

Let u and v be distinct vectors of a vector space V. Show that if $\{u, v\}$ is a basis for V and a and b are nonzero scalars, then both $\{u + v, au\}$ and $\{au, bv\}$ are also bases for V.

Answer:

Since $\{u, v\}$ is a basis for V and each new set has the same size, we only need to check that the new set can generate u and v.

- For $\{u' = u + v,\ v' = au\}$, we have $u = \frac{1}{a} \cdot v'$ and $v = u' - \frac{1}{a} \cdot v'$.
- For $\{u' = au,\ v' = bv\}$, we have $u = \frac{1}{a} \cdot u'$ and $v = \frac{1}{b} \cdot v'$.

Therefore, the two new sets are bases for V.

Problem 4 [10 pts]

Let $v_1, v_2, \ldots, v_k, v$ be vectors in a vector space V, and define $W_1 = \text{Span}(\{v_1, v_2, \ldots, v_k\})$, and $W_2 = \text{Span}(\{v_1, v_2, \ldots, v_k, v\})$.

(a) Find necessary and sufficient conditions on v such that $\dim(W_1) = \dim(W_2)$.

(b) State and prove a relationship involving $\dim(W_1)$ and $\dim(W_2)$ in the case that $\dim(W_1) \neq \dim(W_2)$.

Answer:

(a) $\dim(W_1) = \dim(W_2) \iff v$ can be written as a linear combination of $v_1, \ldots, v_k$. Since $W_1 \subseteq W_2$, we have $\dim(W_1) = \dim(W_2) \iff W_1 = W_2 \iff v \in W_1$, which is equivalent to saying that v can be written as a linear combination of $v_1, \ldots, v_k$.

(b) If $\dim(W_1) \neq \dim(W_2)$, we show that $\dim(W_1) + 1 = \dim(W_2)$. For an arbitrary basis B for W_1, we have $v \notin \mathrm{Span}(B) = W_1$ (otherwise $v \in W_1$ and, by (a), $\dim(W_1) = \dim(W_2)$, contrary to the assumption), so $B \cup \{v\}$ is linearly independent and $\mathrm{Span}(B \cup \{v\}) = \mathrm{Span}(\{v_1, v_2, \ldots, v_k, v\}) = W_2$. Therefore, $\dim(W_1) + 1 = \dim(W_2)$.

Problem 5 [20 pts]

(a) Prove that if W_1 and W_2 are finite-dimensional subspaces of a vector space V, then the subspace $W_1 + W_2$ is finite-dimensional, and $\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$.

Hint: Start with a basis $\{u_1, u_2, \ldots, u_k\}$ for $W_1 \cap W_2$ and extend this set to a basis $\{u_1, u_2, \ldots, u_k, v_1, v_2, \ldots, v_m\}$ for W_1 and to a basis $\{u_1, u_2, \ldots, u_k, w_1, w_2, \ldots, w_p\}$ for W_2.

(b) Let W_1 and W_2 be finite-dimensional subspaces of a vector space V, and let $V = W_1 + W_2$. Deduce that V is the direct sum of W_1 and W_2 if and only if $\dim(V) = \dim(W_1) + \dim(W_2)$.

Answer:

(a) Let $B_0 = \{u_1, u_2, \ldots, u_k\}$ be a basis for $W_1 \cap W_2$. Since B_0 is a linearly independent subset of both W_1 and W_2, we can extend it to a basis $B_1 = \{u_1, u_2, \ldots, u_k, v_1, v_2, \ldots, v_m\}$ for W_1 and to a basis $B_2 = \{u_1, u_2, \ldots, u_k, w_1, w_2, \ldots, w_p\}$ for W_2.

The set $B = \{u_1, u_2, \ldots, u_k, v_1, v_2, \ldots, v_m, w_1, w_2, \ldots, w_p\}$ spans $W_1 + W_2$, since any element of $W_1 + W_2$ can be expressed as the sum of an element of W_1 and an element of W_2, each of which is a linear combination of B_1 and B_2, respectively.

Assume $\sum a_i u_i + \sum b_j v_j + \sum c_l w_l = 0$. Let $x = \sum b_j v_j \in W_1$. Then $-x = \sum a_i u_i + \sum c_l w_l$ is in W_2, which means $x, -x \in W_1 \cap W_2$. Hence $x = \sum d_i u_i$, so $\sum d_i u_i - \sum b_j v_j = 0$, and since B_1 is linearly independent, all d_i and all b_j are zero. The assumed relation then becomes $\sum a_i u_i + \sum c_l w_l = 0$, and since B_2 is linearly independent, all a_i and all c_l are zero as well. Therefore, B is linearly independent.

In conclusion, B is a basis for $W_1 + W_2$, so $W_1 + W_2$ is finite-dimensional and $\dim(W_1 + W_2) = k + m + p = (k + m) + (k + p) - k = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$.

(b) By the definition and (a), we have $V = W_1 \oplus W_2 \iff W_1 \cap W_2 = \{0\} \iff \dim(W_1 \cap W_2) = 0 \iff \dim(V) = \dim(W_1) + \dim(W_2)$.

Problem 6 [8 pts]

Label the following statements as true or false. In each part, V and W are finite-dimensional vector spaces (over $\mathbb{F}$), and T is a function from V to W.

(a) If T is linear, then T preserves sums and scalar products.

(b) If $T(x+y) = T(x) + T(y)$, then T is linear.

(c) T is one-to-one if and only if the only vector x such that $T(x) = 0$ is $x = 0$.

(d) If T is linear, then $T(0_V) = 0_W$.

(e) If T is linear, then $\text{nullity}(T) + \text{rank}(T) = \dim(W)$.

(f) If T is linear, then T carries linearly independent subsets of V onto linearly independent subsets of W.

(g) If $T, U : V \to W$ are both linear and agree on a basis for V, then $T = U$.

(h) Given $x_1, x_2 \in V$ and $y_1, y_2 \in W$, there exists a linear transformation $T : V \to W$ such that $T(x_1) = y_1$ and $T(x_2) = y_2$.

Answer:

The answers are, in order, (a. to d.) T, F, F, T, (e. to h.) F, F, T, F.

For Problems 7-9, prove that T is a linear transformation, and find bases for both $\mathcal{N}(T)$ and $\mathcal{R}(T)$. Then compute the nullity and rank of T, and verify the dimension theorem. Finally, use the appropriate theorems in this section to determine whether T is one-to-one or onto.

Problem 7 [5 pts]

$T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$.

Answer:

For any $x, y \in \mathbb{R}^3$ and $c \in \mathbb{R}$, we have $T(cx + y) = T((cx_1 + y_1, cx_2 + y_2, cx_3 + y_3)) = (c(x_1 - x_2) + (y_1 - y_2), 2cx_3 + 2y_3)$ and $cT(x) + T(y) = c(x_1 - x_2, 2x_3) + (y_1 - y_2, 2y_3) = (c(x_1 - x_2) + (y_1 - y_2), 2cx_3 + 2y_3)$, so $T(cx + y) = cT(x) + T(y)$, which implies T is linear.

$\mathcal{N}(T) = \{(a_1, a_2, a_3) : a_1 = a_2, a_3 = 0\}$ and $\mathcal{R}(T) = \mathbb{R}^2$, so one basis of each is $\{(1, 1, 0)\}$ and $\{(1, 0), (0, 1)\}$, respectively. Therefore, $\text{nullity}(T) = 1, \text{rank}(T) = 2$. Since $\text{nullity}(T) + \text{rank}(T) = 1 + 2 = 3 = \dim(\mathbb{R}^3)$, the dimension theorem is verified.

Since $\mathcal{N}(T) \neq \{0\}$ and $\mathcal{R}(T) = \mathbb{R}^2$, we have that T is not one-to-one but is onto.

Problem 8 [5 pts]

$T : P_2(\mathbb{R}) \to P_3(\mathbb{R})$ defined by $T(f(x)) = xf(x) + f'(x)$.

Problem 8 [5 pts] continued on next page...

Answer:

For any $f(x), g(x) \in P_2(\mathbb{R})$ and $c \in \mathbb{R}$, we have $T(cf(x) + g(x)) = x(cf(x) + g(x)) + cf'(x) + g'(x)$ and $cT(f(x)) + T(g(x)) = cxf(x) + cf'(x) + xg(x) + g'(x)$, so $T(cf + g) = cT(f) + T(g)$, which implies T is linear.

For $f(x) = a_0 + a_1x + a_2x^2$, we have $T(f(x)) = a_1 + (a_0 + 2a_2)x + a_1x^2 + a_2x^3$, then $\mathcal{N}(T) = \{0\}$ and $\mathcal{R}(T) = \{a_0 + a_1x + a_2x^2 + a_3x^3 : a_0 = a_2\}$, so one basis of each is $\emptyset$ and $\{1 + x^2, x, x^3\}$, respectively. Therefore, $\text{nullity}(T) = 0, \text{rank}(T) = 3$. Since $\text{nullity}(T) + \text{rank}(T) = 0 + 3 = 3 = \dim(P_2(\mathbb{R}))$, the dimension theorem is verified.

Since $\mathcal{N}(T) = \{0\}$ and $\mathcal{R}(T) \neq P_3(\mathbb{R})$, we have that T is one-to-one but is not onto.

Problem 9 [5 pts]

$T : M_{n\times n}(\mathbb{F}) \to \mathbb{F}$ defined by $T(A) = \text{tr}(A)$. Recall that

$$\text{tr}(A) = \sum_{i=1}^{n} A_{ii}.$$

Answer:

For elements in $M_{n\times n}(\mathbb{F})$, the addition and scalar multiplication affect the trace directly, so T is linear: for $A, B \in M_{n\times n}(\mathbb{F})$ and $c \in \mathbb{F}$, we have $\text{tr}(cA + B) = c\,\text{tr}(A) + \text{tr}(B)$.

$\mathcal{N}(T) = \{A \in M_{n\times n}(\mathbb{F}) : \sum_{i=1}^{n} A_{i,i} = 0\}$ and $\mathcal{R}(T) = \mathbb{F}$, so one basis of each is $\{E^{ij} : 1 \leq i, j \leq n, i \neq j\} \cup \{E^{ii} - E^{nn} : 1 \leq i < n\}$ and $\{1\}$, respectively, where E^{ij} denotes the matrix whose only nonzero entry is 1 in the ith row and jth column. Therefore, $\text{nullity}(T) = n^2 - 1, \text{rank}(T) = 1$. Since $\text{nullity}(T) + \text{rank}(T) = n^2 - 1 + 1 = n^2 = \dim(M_{n\times n}(\mathbb{F}))$, the dimension theorem is verified.

For $n = 1$, since $\mathcal{N}(T) = \{0\}$ and $\mathcal{R}(T) = \mathbb{F}$, we have that T is one-to-one and onto. For $n \geq 2$, since $\mathcal{N}(T) \neq \{0\}$ and $\mathcal{R}(T) = \mathbb{F}$, we have that T is not one-to-one but is onto.

Problem 10 [10 pts]

In this exercise, $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a function. For each of the following parts, state why T is not linear.

(a) $T(a_1, a_2) = (1, a_2)$

(b) $T(a_1, a_2) = (a_1, a_1^2)$

Answer:

(a) $2 \cdot T((0, 0)) = 2 \cdot (1, 0) = (2, 0)$ while $T(2 \cdot (0, 0)) = T((0, 0)) = (1, 0)$.

(b) $2 \cdot T((2, 2)) = 2 \cdot (2, 4) = (4, 8)$ while $T(2 \cdot (2, 2)) = T((4, 4)) = (4, 16)$.

Problem 11 [10 pts]

Prove that there exists a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^3$ such that $T(1,1) = (1,0,2)$ and $T(2,3) = (1,-1,4)$. What is $T(8,11)$?

Answer:

The linear transformation is $T(x,y) = ((3x-2y)+(y-x), -(y-x), 2(3x-2y)+4(y-x)) = (2x-y, x-y, 2x)$, with all operations linear. Then $T(8,11) = (5,-3,16)$.

Problem 12 [10 pts]

Let $T : P(\mathbb{R}) \to P(\mathbb{R})$ be defined by $T(f(x)) = f'(x)$. Recall that T is linear. Prove that T is onto, but not one-to-one.

Answer:

We have $[x^n]f'(x) = (n+1)[x^{n+1}]f(x)$, which directly implies $\mathcal{N}(T) = \{c : c \in \mathbb{R}\} \neq \{0\}$ and $\mathcal{R}(T) = P(\mathbb{R})$. Therefore, T is onto but not one-to-one.

Problem 13 [10 pts]

Let V and W be finite-dimensional vector spaces and $T : V \to W$ be linear.

(a) Prove that if $\dim(V) < \dim(W)$, then T cannot be onto.

(b) Prove that if $\dim(V) > \dim(W)$, then T cannot be one-to-one.

Answer:

(a) We know $\mathcal{R}(T)$ is a subspace of W, and $\text{rank}(T) \leq \dim(V) < \dim(W)$ by the dimension theorem, so $\mathcal{R}(T) \subsetneq W$. Therefore, T cannot be onto.

(b) We know $\mathcal{R}(T)$ is a subspace of W, and $\text{rank}(T) \leq \dim(W) < \dim(V)$, so by the dimension theorem $\text{nullity}(T) = \dim(V) - \text{rank}(T) \geq 1$ and $\mathcal{N}(T) \neq \{0\}$. Therefore, T cannot be one-to-one.

The following definitions are used in Problems 14, 15.

Definition 1

Let V be a vector space, and let $T : V \to V$ be linear. A subspace W of V is said to be **T-invariant** if $T(x) \in W$ for every $x \in W$, that is, $T(W) \subseteq W$. If W is T-invariant, we define the **restriction of T on W** to be the function $T_W : W \to W$ defined by $T_W(x) = T(x)$ for all $x \in W$.

Problems 14, 15 assume that W is a subspace of a vector space V and that $T : V \to V$ is linear.

Problem 14 [8 pts]

Prove that the subspaces $\{0\}$, V, $\mathcal{R}(T)$, and $\mathcal{N}(T)$ are all T-invariant.

Answer:

- Since T is linear, we have $T(0) = 0$, so $\{0\}$ is T-invariant.
- Since T is $V \to V$, we have $T(V) \subseteq V$, so V is T-invariant.
- Since $T(S) \subseteq \mathcal{R}(T)$ for any set $S \subseteq V$, we have $T(\mathcal{R}(T)) \subseteq \mathcal{R}(T)$, so $\mathcal{R}(T)$ is T-invariant.
- Since $T(\mathcal{N}(T)) = \{0\} \subseteq \mathcal{N}(T)$, we have $\mathcal{N}(T)$ is T-invariant.

Problem 15 [8 pts]

If W is T-invariant, prove that T_W is linear.

Answer:

For any $x, y \in W$ and $c \in \mathbb{R}$, we have $cx + y \in W$ since W is a vector space, and $T(x), T(y) \in W$ since W is T-invariant. Therefore, $T_W(cx + y) = T(cx + y) = cT(x) + T(y) = cT_W(x) + T_W(y)$, so T_W is linear.

Problem 16 [12 pts]

A function $T : V \to W$ between vector spaces V and W is called **additive** if $T(x + y) = T(x) + T(y)$ for all $x, y \in V$. Prove that if V and W are vector spaces over the field of rational numbers, then any additive function from V into W is a linear transformation.

Answer:

We only need to verify the scalar multiplication. For any $x \in V$ and $n \in \mathbb{N}$, we have $T(nx) = T((n-1)x) + T(x) = \cdots = nT(x)$ by induction, then $T(x) = \frac{1}{n}T(nx)$. For any $x \in V$, $c = \frac{p}{q} \in \mathbb{Q}$ and $p, q \in \mathbb{N}$, we have $T(cx) = T(\frac{p}{q}x) = \frac{1}{q}T(px) = \frac{p}{q}T(x) = cT(x)$ by the previous relation since $\frac{1}{q}x \in V$. For $c = 0$ and $c < 0$, we have $T(0) = T(x + (-x)) = T(x) + T(-x) = 0$ and $T(-x) = -T(x)$, so $T(cx) = cT(x)$ holds as well.

Therefore, any additive function T over $\mathbb{Q}$ is linear.

The following exercise requires familiarity with the definition of quotient space given in Problem 11 of homework 2.

Problem 17 [10 pts]

Let V be a vector space and W be a subspace of V. Define the mapping $\eta : V \to V/W$ by $\eta(v) = v + W$ for $v \in V$.

(a) Prove that η is a linear transformation from V onto V/W and that $\mathcal{N}(\eta) = W$.

Problem 17 [10 pts] continued on next page…

(b) Suppose that V is finite-dimensional. Use (a) and the dimension theorem to derive a formula relating $\dim(V)$, $\dim(W)$, and $\dim(V/W)$.

Answer:

(a) For any $x, y \in V$ and $c \in \mathbb{F}$, we have $\eta(cx+y) = (cx+y)+W$ and $c\eta(x)+\eta(y) = c(x+W)+(y+W) = (cx+y)+W$ by the properties we proved in Hw2 P11. Therefore, η is linear, and it is onto since every element of V/W is $v+W = \eta(v)$.

We know $v_1 + W = v_2 + W \iff v_1 - v_2 \in W$, so $v + W = 0 + W \iff v \in W$, which implies $\mathcal{N}(\eta) = W$.

(b) We know $\mathcal{R}(\eta) = V/W$, so by the dimension theorem, we have $\dim(V) = \text{nullity}(\eta) + \text{rank}(\eta) = \dim(W) + \dim(V/W)$.

Problem 18 [16 pts]

Let $\alpha_1, \alpha_2, \cdots, \alpha_m$ $(m \geq 2)$ be vectors in a vector space over $\mathbb{R}$. α_m is not a zero vector. Prove that the following statements are equivalent:

1. for any real numbers $k_1, k_2, \cdots, k_{m-1}$, the vectors

$$\beta_1 = \alpha_1 + k_1\alpha_m,\ \beta_2 = \alpha_2 + k_2\alpha_m,\ \cdots,\ \beta_{m-1} = \alpha_{m-1} + k_{m-1}\alpha_m$$

are linearly independent.

2. The vectors $\alpha_1, \alpha_2, \cdots, \alpha_m$ are linearly independent.

Answer:

$\Longrightarrow$: Assume $\sum_{i=1}^{m} a_i\alpha_i = 0$, then for any $k_{1\ldots m-1}$ such that $\sum_{i=1}^{m-1} a_i k_i = a_m$, we have $\sum_{i=1}^{m-1} a_i\beta_i = 0$. Since $\beta_{1\ldots m-1}$ are linear independent, we have $a_i = 0$ for all $1 \leq i < m$, and since α_m is nonzero, a_m is 0 as well for the zero sum. Therefore, $\alpha_1, \alpha_2, \cdots, \alpha_m$ are linearly independent.

$\Longleftarrow$: For any $k_{1\ldots m-1}$, assume $\sum_{i=1}^{m-1} b_i\beta_i = 0$, then $\sum_{i=1}^{m-1} b_i\alpha_i + \sum_{i=1}^{m-1} b_i k_i\alpha_m = 0$. Since $\alpha_{1\ldots m}$ are linear independent, we have $b_i = 0$ for all $1 \leq i < m$. Therefore, $\beta_1, \beta_2, \cdots, \beta_{m-1}$ are linearly independent for any $k_{1\ldots m-1}$.